Challenge Problems

7. The average marks of six students is 8. One more student mark is added and the mean is still 8. Find the student mark that has been added.

8. Calculate the mean, mode and median for the following data:
22, 15, 10, 10, 24, 21.

9. Find the median of the given data: 14, −3, 0, −2, −8, 13, −1, 7.

10. Find the mean of first 10 prime numbers and first 10 composite numbers.

Summary

- Based on the purpose, appropriate data has to be collected and organised to find the representative of data.
- Representative of data are also known as measures of central tendency.
- Arithmetic mean is the most commonly used representative of data and is calculated by the formula.

  $$\text{Arithmetic Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

- Mode is the value of the data which occurs maximum number of times.
- A data may have more than one mode as well as no mode.
- A data is of large size, mode can be found out after grouping.
- Median is the middle most value of the given data.
- To find the median for the given data,
  - (i) arrange the data in ascending or descending order.
  - (ii) If the number of terms (n) is odd, then $\left(\frac{n+1}{2}\right)^{th}$ term is the median.
  - (iii) If the number of terms (n) is even, then average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2}+1\right)^{th}$ terms is the median.

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